Part 1: Use Case Diagram – Parking System

System Overview

The Parking System manages parking lot access, customer registration, permit purchases, and enforcement. It supports customers parking vehicles, the parking office managing permits and enforcement, and the parking lot infrastructure tracking vehicle entry and exit.

Key Actors

- Customer – A person who parks a vehicle and may purchase a parking permit
- Parking Office – Administrative staff managing permits, pricing, and enforcement
- Parking Lot System – Automated system controlling gates, sensors, and availability
- Enforcement Officer – Monitors compliance and issues citations
- Payment System – An external system that processes payments

Key Interactions

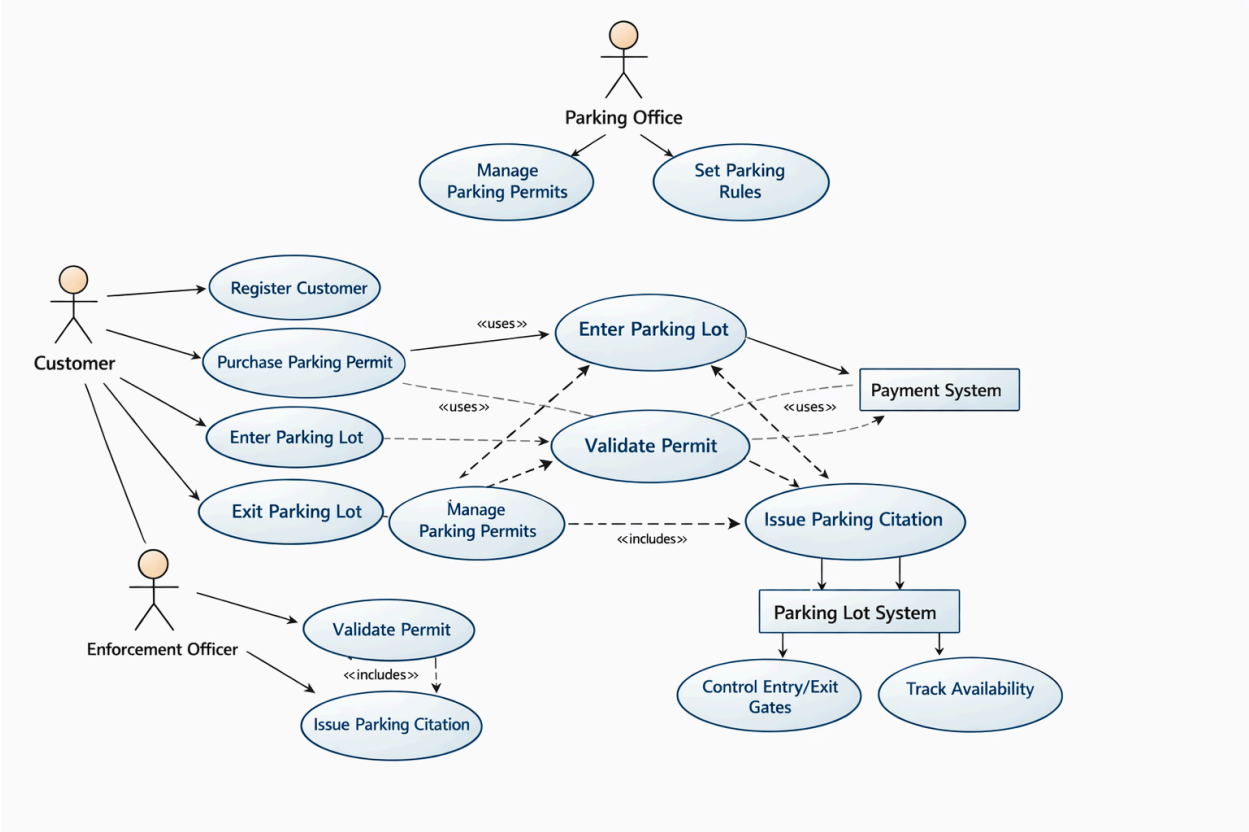
- Customer drives into the parking lot
- Customer registers an account
- Customer purchases a parking permit
- Parking system validates permit
- Parking system allows or denies entry
- Enforcement officer checks parked vehicles
- Parking office manages permits and pricing

Major Use Cases

- Register Customer
- Purchase Parking Permit
- Enter Parking Lot

- Exit Parking Lot
- Validate Permit
- Issue Parking Citation
- Manage Parking Permits
- Process Payment

Parking Use Case Diagram.



Part 2: Individual Use Case Documentation

Use Case Title

Purchase Parking Permit

Primary Actor

Customer

Secondary Actors

- Parking Office
- Payment System

Preconditions

- Customer has access to the parking system (online or kiosk)
- Parking permits are available for purchase
- The payment system is operational

Postconditions

- Success:
    - A parking permit is issued and associated with the customer's vehicle
    - Payment is successfully processed
- Failure:
    - The permit is not issued
    - Customer is notified of the issue

Normal Flow

1. Customer accesses the parking system.

2. The customer selects the option to purchase a parking permit.
3. System prompts the customer to enter vehicle and personal details.
4. Customer selects permit type (daily, monthly, annual).
5. System displays pricing and terms.
6. Customer confirms the purchase.
7. The system sends payment details to the payment system.
8. The payment system approves the transaction.
9. The parking system issues the permit and stores it in the system.
10. Customer receives confirmation of the permit purchase.

Alternative Flows

- Payment Failure:
    - Payment system declines the transaction.
    - Customer is notified and prompted to retry or cancel.
- Invalid Information:
    - Customer enters incorrect vehicle information.
    - System prompts customer to correct the details.

Failure Conditions

- Payment system unavailable
- Invalid customer or vehicle data
- System error during permit creation

Reflection

Completing this assignment was a valuable exercise in understanding how system behavior can be modeled from a user-focused perspective. One of the easier aspects was identifying the key

actors and high-level interactions. Parking systems are familiar to most people, so imagining how customers, parking staff, and enforcement officers interact with the system felt intuitive.

The most challenging part was deciding how much detail to include without overcomplicating the use cases. It was tempting to add too many technical steps or edge cases, but the assignment emphasized focusing on critical system behaviors rather than implementation-level details. Keeping the use cases clear and concise required careful judgment.

What helped the most was following the structured use case template. Breaking the use case into actors, preconditions, normal flows, and alternative flows made the process manageable and logical. It also ensured that the use case remained user-centered instead of system-centered. Reviewing example use case diagrams and written use cases from class materials further clarified expectations.

One thing I wish I had known earlier was how important it is to think in terms of goals rather than actions. For example, “Purchase Parking Permit” represents a user goal, while actions like entering data or processing payment support that goal. Recognizing this distinction made it easier to design meaningful use cases.

Several implementation decisions were made intentionally. I chose to include a Payment System as a separate actor to reflect real-world integration with external services. I also separated enforcement responsibilities from the parking office to better represent distinct roles within the system. These decisions help keep the model realistic while maintaining clarity.

Overall, this assignment improved my understanding of how use cases help bridge the gap between user needs and system design, and it reinforced the importance of clear documentation early in the development process.